Quality Assurance(QA)

It is defined as activity to ensure that an organization is providing the best possible products and services to customers.

4 steps of QA cycle:

- Plan - Identify goals and baseline
- Do - Organize supporting documentation,procedures,training materials in a document management system
- Check - Control,measure and monitor outputs to ensure they meet expected criteria
- Act - Review the findings of quality management process

QA Plan

Development
Baseline System
Production
Release to market

Software Development Life Cycle(SDLC)

It is a process of developing a software industry to develop,design,and test high quality software.It is a detailed plan to develop,maintain and replace and alter and enhance specific software.

Requirement analysis
Software Design
Software build
Testing
Deployment
Maintenance

Requirement Analysis

Requirement Analysis is the most important and necessary stage in SDLC.Planning for the quality assurance requirements and identifications of the risks associated with the projects is also done at this stage.

Software Design

The next phase is about to bring down all the knowledge of requirements, analysis, and design of the software project. This phase is the product of the last two, like inputs from the customer and requirement gathering

Software Build

The implementation of design begins concerning writing code. Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

Testing

The code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage.During this stage, unit testing, integration testing, system testing, acceptance testing are done.

Deployment

Once the software is certified, and no bugs or errors are stated, then it is deployed.Then based on the assessment, the software may be released as it is or with suggested enhancement in the object segment.After the software is deployed, then its maintenance begins.

Maintenance

Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time.This procedure where the care is taken for the developed product is known as maintenance.

SDLC Models

Waterfall Model

The waterfall model is a continuous software development model in which development is seen as flowing steadily downwards (like a waterfall) through the steps of requirements analysis, design, implementation, testing (validation), integration, and maintenance.

RAD Model

RAD or Rapid Application Development process is an adoption of the waterfall model; it targets developing software in a short period. The RAD model is based on the concept that a better system can be developed in less time by using focus groups to gather system requirements.

Business Modeling
Data Modeling
Process Modeling
Application Generation
Testing and Turnover

Spiral Model

The spiral model is a risk-driven process model. This SDLC model helps the group to adopt elements of one or more process models like a waterfall, incremental, waterfall, etc. The spiral technique is a combination of rapid prototyping and concurrency in design and development activities.the cycle is to evaluate these different alternatives based on the objectives and constraints. The focus of evaluation in this step is based on the risk perception for the project.

V Model

In this type of SDLC model testing and the development, the step is planned in parallel. So, there are verification phases on the side and the validation phase on the other side. V-Model joins by Coding phase.

Incremental Model

It is necessarily a series of waterfall cycles. The requirements are divided into groups at the start of the project. For each group, the SDLC model is followed to develop software. The SDLC process is repeated, with each release adding more functionality until all requirements are met.

Agile Model

Agile methodology is a practice which promotes continued interaction of development and testing during the SDLC process of any project. In the Agile method, the entire project is divided into small incremental builds. All of these builds are provided in iterations, and each iteration lasts from one to three weeks.

Software Testing Life Cycle(STLC)

The process of testing a software in a well planned & systematic way.

Requirement Analysis
Test Planning
Test case Development
Test Environment Setup
Test Execution
Test closure

Comparison of SDLC and STLC

SDLC	STLC
Besides development other phases like testing is also included.	It focuses only on testing the software.
Creation of reusable software systems is the end result of SDLC.	A tested software system is the end result of STLC.
Goal of SDLC is to complete successful development of software.	STLC phases are performed after SDLC phases.
In SDLC, the development team makes the plans and designs based on the requirements.	In STLC, the testing team(Test Lead or Test Architect) makes the plans and designs.
SDLC involves a total of six phases or steps.	STLC involves only five phases or steps.

Defect metrics and Defect Management Process

Defect Management is a systematic process to identify and fix bugs.

- Discovery of defect
- Defect Categorization
- Fixing of defect developers

- Verification of testers
- Defect closure
- Defect Reports at the end of the projects

Importance of Testing:

Testing is executing a system in order to identify any gaps, errors, or missing requirements in contrast to the actual requirements.

Automation Testing :

It is the type of testing in which we can use the tools(automation) for the testing.It is faster than manual because it is done by the tools.Automation Testing is used to re-run the test scenarios that were performed manually, quickly, and repeatedly.

Manual Testing:

Manual testing includes testing a software manually, i.e., without using any automated tool or any script. In this type, the tester takes over the role of an end-user and tests the software to identify any unexpected behavior or bug. There are different stages for manual testing such as unit testing, integration testing, system testing, and user acceptance testing.

Black box testing

It is the process of checking the functionality of an application as per the customer requirements externally

White box Testing

software testing technique that tests the software by using the knowledge of internal data structures, physical logic flow, and architecture at the level of source code.

Gray Box Testing

The gray-box testing involves inputs and outputs of a program for the testing purpose but test design is tested by using the information about the code.

Functional testing

Unit Testing

It is the first level of testing of any software product.It is the process of testing the module of an application independently or test all the functionality is known as unit testing.

Integration Testing

It is a type of software testing in which each unit,modules or components are tested as a combined entity.

Smoke & Sanity Testing

Smoke : It is one of the popular software testing services that perform after build to find if the critical functionalities of the application are working fine or not. Designing and executing basic functionalities of the application.

eg: Login Page

Sanity: Sanity test is narrow and deep and unscripted.It is narrow regression test that focuses on a very few functionalities.It is used to determine if the section of the application is still working after a minor change.

Regression & Retesting

Retest: It is Executing previously failed tests against new software to check if problem is solved.After a defect has been fixed, re-testing is performed to check the scenario under the same environmental conditions.

Regression Testing: It is a Black box testing technique that consists of re-executing those tests that are impacted by code changes.Mostly it is changed by the following cases:

- Change in requirement
- New functionality added to the software
- Performance issue fix
- Environment change

System testing

It is a process of checking end-to-end applications or the software for the user is known as System testing. In this it will navigate through all the necessary modules of an application and check all the end features or end business works fine and test a product as a whole system.

User Acceptance Testing

It is used to determine whether a software system has met the system requirements.It's main purpose is to evaluate the system with the compliance with business requirements and meet its requirement criteria for delivery to end users.

Non-Functional Testing

Non-functional testing assesses application properties that aren't critical to functionality but contribute to the end-user experience.

Performance Testing

Performance testing is a well -structured and clear specification about expected speed must be defined.It eliminates the reason behind the slow and limited performance of the software.

Security Testing

Security testing is used to detect the security flaws of the software application. The testing is done via investigating system architecture and the mindset of an attacker. Test cases are conducted by finding areas of code where an attack is most likely to happen.

Portability

The portability testing of the software is used to verify whether the system can run on different operating systems without encountering any bug. This test also tests the working of software when there is the same operating system but different hardware.

Reliability testing

Reliability test assumes that the software system is running without fail under specified conditions or not. The system must be run for a specific time and number of processes. If the system fails under these specified conditions, a reliability test will be failed.

Testcase

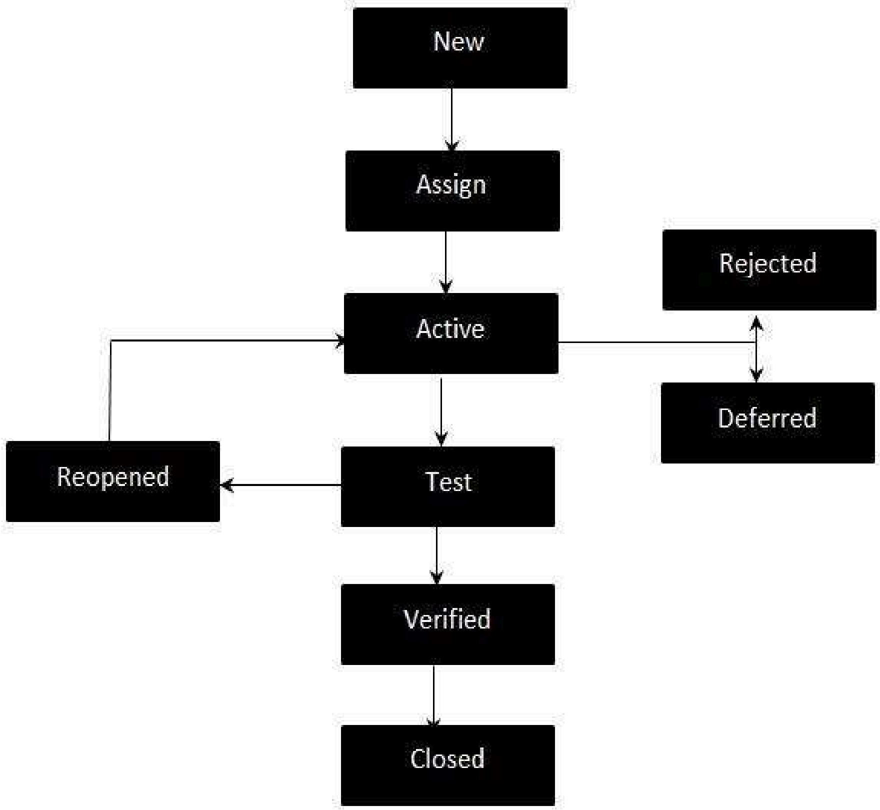
A test case is a set of actions performed on a system to determine if it satisfies software requirements and functions correctly.

Testcase Parameters:

- Test Case ID
- Precondition
- Test Case Scenario
- Test Case Name
- Test Case Steps
- Priority
- Expected Result
- Actual Result
- Status

BUG Life Cycle:

It is also known as the Bug Life Cycle.It varies from organization to organization and also from project to project as it is governed by the software testing process and also depends upon the tools used.



Bug Life Cycle States:

- New - Potential defect that is raised and yet to be validated
- Assigned - Assigned against a development team to address it but not yet resolved.
- Active - The Defect is being addressed by the developer and investigation is under progress. At this stage there are two possible outcomes; viz - Deferred or Rejected.
- Test - The Defect is fixed and ready for testing.
- Verified - The Defect that is retested and the test has been verified by QA.
- Closed - The final state of the defect that can be closed after the QA retesting or can be closed if the defect is duplicate or considered as NOT a defect.
- Reopened - When the defect is NOT fixed, QA reopens/reactivates the defect.
- Deferred - When a defect cannot be addressed in that particular cycle it is deferred to future release.

- Rejected - A defect can be rejected for any of the 3 reasons; viz - duplicate defect, NOT a Defect, Non Reproducible.

Agile Process

Agile software development refers to software development methodologies centered around the idea of iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams. The ultimate value in Agile development is that it enables teams to deliver value faster, with greater quality and predictability, and greater aptitude to respond to change. Scrum and Kanban are two of the most widely used Agile methodologies.

Scrum

A “process framework” is a particular set of practices that must be followed in order for a process to be consistent with the framework.

STLC:-

- **STLC is** a process of evaluating the functionality of a software application to find software bugs
- It checks if the software met the specific requirement and identifies any defect in the software.
- Verifying and validating the software product
- Checks that the functionality works as per the user requirement

Requirement analysis:-

- Qa teams understand the requirements
- Qa team study the requirement for the testing point of view
- Qa team interacts with various stakeholders
- Requirements are either functional and non - functional

Testing plan:-

- It helps to determine the testing
- Testing strategy is defined in the document
- Test manager determines the effort and cost estimate for the project

Test case:-

- Testing team develops the test case for testing
- Also developed the test data if required for testing

Test environment setup;-

- Set of hardware and software for the testing team to execute test case

Test Execution:-

- If test case id fail the it can be reported to the development team
- Identify the bug

Test closure report:-

- Test summary
- Identifiers
- Test summary
- Summary of result
- Evaluation
- Summary of activities

Test case:-

A test case is a document, which has a set of test data, test scenario, test case description, test steps, expected result, Actual result and the test status.

Typical Test Case Parameters:-

1. Test Case ID – Unique id to the test case.
2. Test Scenario – Functionality of the specific test case.
3. Test Case Description – detailed explanation of the test scenario.
4. Test Steps – step by step explanation of the test case.
5. Test Data –
6. Expected Result – the ideal result that should be achieved once a test case has been executed
7. Actual Result - Actual Outcome also known as actual result, which a tester gets after performing the test.
8. Test Status – pass / fail

Waterfall model:-   (fixed model)

- Old and traditional model
- Linear model
- Comes under sdlc

Requirement analysis:-

- Documentation (requirement cannot be changed)
- Collects the requirements from the customers and prepare a SRS document (software requirement specification document)

System design:-

- Based on the document
- Designer design the software as high level module and low level module

Implementation:-

- Based on the design the developers implement the software(development)

Testing:-

-test the software

Deployment:-

- Deploy the software to customer
- Customer will use the software

- Advantage :-
    - quality will be good (because of detailed document)
    - long term product

- requirement changes are not allowed chances of finding bug is less
- initial investment is less
- prefered for small project

-disadvantages:-

- requirement changes are not allowed
- if there is defect in the requirement that will be continued till the last phase
- total investment is more (if we do some rework)
- testing will start only coding (only one phase of testing)

V model / V-V - model (verification and validation model)

- Software development process model
- Verification part
- - BRS/CRS/URS (Business Requirement specification(business unit - huge document) / customer requirement specification / user requirement specification) - based on the testing user acceptance testing will be conducted based on the document. -
- SRS (Software requirement specification document)- understand by technical people - prepared by PM
- HLD/LLD (High lived design document / low level design document) - prepared by designers -
- Coding - development will start

- Validation part -
- Unit testing - testing the single component (developers) - white box testing
- Integration testing - combine 2 module and testing the main module (developers) white box testing
- System testing - testing the product is working bases on custer requirement (testers) - black box testing
- User acceptance testing - customers will test along with testers (testers) - black box testing
- Every phase testing will be done

Agile Process:- (2 to 3 week)

- Combination of iterative(repeating same process) and incremental (adding new features) process models.
- Principle - deliver the piece of software with few functionality. (eg customer expectation 100 features in software but in agile process - not deliver one single software with 100 features - develope 5 to 10 features and test it and release the small piece of software to the customer)
- Accept the requirement changes at any time of development .
- Good communication between developer and customer , testers , business analyst

Advantages :-

- Requirement changes are allowed
- Releases will be very faster (2 -3 week)
- Easy model

Disadvantage:-

- Less and no documentation - because releases are faster.

SCRUM:-

- Framework through which we develop and test the software and finally release to the customer.
- To follow the agile principles scrum framework is helpful (say how to follow)

- Set of people will be involved - (scrum team / agile team - product owner, scrum master, development team and testing team) 5 to 9 numbers
- Product owner - define the features of the product (in form of user story and epic)- contact the customer - collect all the functionality from the customer and then he will arrange in priority wise - he can reject and accept the features.
- Scrum master - not a developer not a tester - roll is facilitating (awareness) and driving the agile process - meet organizer
- Developer / testing - developing and testing the software.

Scrum terminology :-
- User story - feature / module / software.
- Epic - collection of user stories.
- Product backlog (document / excel sheet) - list of user story user stories prepared by the product owner at the beginning of the process.
- Sprint - a period of time to complete a user stories (design.testing) - product owner
- Sprint planning meeting - meet conduct with a agile team to define what can be delivered in the sprint (how many stories are there in sprint - how many stories we develop and test in that sprint - what is the duration of the sprint)
- Sprint backlog - list of stories to be executed in the sprint (only the few number of stories accepted by dev and QA) - subset of product backlog
- Scrum meeting - every day 15min of scrum call / meeting - scrum master - with the scrum team - update the status
- Sprint retrospective meeting - final meet after completing the sprint - scrum team

QA process

Process of QA:-
1 QA initiation
2 QA planning
3 QA Execution stage
4 QA reporting and analyzing stage

QA initiation stage
- Customer business / use case requirement analysis - QA team interacts with pm /sw or the customer to know about the requirements of the project
- PRD review and analysis - QA reviews the PRD meet with pm/sw and understand the detail requirement of the project.
- SDD review and analysis - QA team reviews the add team to understand the software implementation and test approach required for the project.

QA planning
- Detail test case development,review and approval process - Team prepares detailed test case with expected results and reviewed by qa team , sw team and customer.
- Test environment setup - (test setup, device under test , devices required for the complete test case execution process) - tools readlines - (Hardware and software tool setup (for performance and security testing) and test case capture tool).
- Test application - QA team identify test application
- Test execution plan - Developed by qa lead based on the project test requirement.

QA execution :-

- Detail test case execution - test cases are executed as per the requirement analysis on the available tools.
- Defect logging - identified defect will be filed as per the process
- Bug verification / retest process - Bug fix release image should be retested to ensure the bug is fixed properly.

QA report & analysis :-

- Test execution report - Generated with the X-ray tool with the test cases with pass or fail reports
- Defect summary report - Focuses on the total bug and the functionality impact on priority base.
- Bug fix matric analysis - o be done at every stage
- Project assessment report - New / enhanced report are shared over email to QA head , BU head and the stakeholders.
- Customer test report - Customer review and approval.

PR016 - QUALITY ASSURANCE TESTING PROCESS
TP022 - TEST PLAN TEMPLATE
TP023 - TEST DEFECT LOG TEMPLATE

Mobile app testing

1) Android vs Ios

Android	IOS
Has back button	Has only one button
Has google assistant	Has siri as voice assistance
Available more than 100+ languages	Has only 34 languages
Based on users requirement	Difficult to use

Mobile application :-

App crashes / does not work - e.g. sale for one plus will start at midnight - traffic is so much and the mobile applications can't handle it. We design the mobile application with a particular load when 100 thousand users using the application at the same time simultaneously the app crashes.

How to install application on mobile phone:-(without using app store etc)

Extension for android - .apk
Extension for ios - .ip

- In android phone - install application - plug the usb cable - transfer the application to file manager - click on apk it can be installed.
- 1st screen comes on opening the application is splash screen
- Three dots - settings
- Hamburger icon - main menu

How to test the mobile application?

1. Installing the application successfully
2. Supports which minimum os which the application supports (eg marshmallow android 6 version) id installed below os version 6 it cannot be supported.it wont word as excepted.

Web application testing

HTTP - HyperText Transfer Protocol

- Used for data transfer
- But the data exchange is not secure
- Files exchanged using the http goes as plain text
- Default port 80

HTTPS - HyperText Transfer Protocol Secure

- More secure than http
- Https = http + cryptographic protocols
- Default port 443

HTTP	HTTPS
HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Url begins with “http://”	Url begins with “https://”
insecure	secure
Works on application layer	Works on transport layer
Does not need any certificate	Needs SSL certificate that provides the encryptions of the data
Page loading speed is fast	Page loading speed is low because of the additional features like security.

Web app testing techniques:-

- Boundary value analysis - Testing the field with the user requirement with 0 to 5000 should accept that is the correct boundary range
- How to test this scenario - num less than 0 and greater than 5000 is invalid.

Need for web application testing:-

- Testing websites content are meaningful
- Also test the advertisement are relevant to that app
- User friendly
- Browser compatibility (should run in all the browsers in all the platforms)
- Reliability (crash free, work with internet is slow, secure or not, malicious activity should not be performed)
- Max num of users accessibility.
- Secured access of the application (https)(datas should not be hacked)

Working of a website;-

- Client (local computer, pc,laptop) - http request - internet - web server - database - html response - user

Web server vs web page;-

- Web server serves the web page
- Web page request the web server

Local machine - contains the data
Web browser - requests - web server - fetch the data from local machine

Web Server	Web Browser
It is a software that provides documents that are requested by the web browsers.	Browser sends the http request and the web server receives the request and responds.
Data storage- database (remote database)	Data storage - local storage in the form of cookies.
Web server can be installed with internet access (network should be same)	Web browser should be installed in local machine

Common errors in web application:-

- See symbol of error not the error itself
- Web app configuration - every website version controlling system(trace the changes happened in the version eg of the code has deleted)

Types of websites to be tested:-

- Static website - each page in this website is saved as single html file (faster- response term is faster and secure)
- Dynamic website - store into different html files
- E-commerce website
- Mobile website

Web app testing techniques;-

- Usability - how much user friendly the application - how easy to use this application - how many different kind of activity can be performed
- Compatibility - application is working on all the types if windows - working on low quality and also the high quality mobile

Functional testing:-

- Link testing - medium through navigate between other screen.(pathway)

    - internal link- hyperlink on a web page to another page ie)image,document.
    -outgoing link - link the point from our one website to another
    -anchor link - helps to link the content of the same page that has an anchor attached; it is an unique id attached to the content block.
    -mail to links - clickable link that automatically opens a new mail in the reader default email client such as outlook or email.

Usability testing :-

- Usability - how much user friendly the application - how easy to use this application - how many different kind of activity can be performed
- Navigating testing of the webpage
- Content of the web page
- User requirements

Interface testing:-(data flow)

- Server side data flow -
- Database server

Compatibility testing:-

Compatibility - application is working on all the types if windows - working on low quality and also the high quality mobile

- Web application is running the same in all the browsers.
- B to B application - business to business - specific company (vvdn) - (in vvdn application is running in chrome the all the user can use chrome os compatibility testing is not needed)
- B to C - business to client - amazon (customers will use this application in different browsers so the compatibility testing should be done)
- Commonly used browsers and the operating system and the devices like mobile, laptop and smartphones.

Performance testing:-

- Stability and response time (how fast the response time)
- Types

- load testing - if the app is customize for 50 user if it is b to b customer then only the 50 user will be using the particular web site .if all the user is hitting the url it should be same for all the user response time should be same,
- stress testing - is designed to access at 50 user max then testing 100 users the response time will be differ.(testing the break point by applying more number of load)
- scalability testing - applying the load more than the desired number at which point it's going to crash.
- soak testing / endurance testing .- continues load for a longer period - will it crash or run smoothly.

Security testing:-

- Url manipulation - copying the logged url to another tab should not get login directly.
- Session expiry - if we are idle while filling the application it gets session expired (banking application)

- Sql injection - ul — application layer – db — back to ui

Hackers - commands directly - with user name directly hits the particular data.

- Cookies based -

Session cookies - on particular time data will be deleted automatically once the session expires.
Persistent cookies - data will be stored user has to delete nd get deleted in particular

- Cross site request forgery - testing the url is encrypted

Eg banking application - should not reflect in url.

- Cross site scripting - pop up appears should be ignored from any other site (we should nt click on the pop up while entering the confidential data like banking details)

Desktop application testing

- Desktop applications are working with the fullest potential (eg-calculator, alarm etc..)
- Compared to web application bugs are lesser in desktop applications.
- Mainly used for gaming applications.

Techniques for desktop application testing:-
- Internet availability
- Different hardware applications - (i3,i5 etc) - high end games cannot be played in low end configuration.
- Variant in os - application should work fine il all os
- Security of the application - application should be secured and encrypted.

Code complicity - how difficult the code is.

Requirement analysis and writing test case
Black box testing.
Boundary value analysis
- password min 8 max 10
- Valid class - input value between 8 to 10
- Invalid class -if less than 8 and greater than 10

Equivalence class partitioning
- Based on the behavior we keep the input in one category
- Driving classes based on the behavior on the input

Load testing tools:
Jmeter:-

Load testing:-
- if the app is customize for 50 user if it is b to b customer then only the 50 user will be using the particular web site .If all the user is hitting the url it should be same for all the user response time should be same

Volume tasting:-
- Fetching the high volume of data from database (testing the time taking to fth the data)

Stress testing:-

- is designed to access at 50 user max then testing 100 users the response time will be differ.(testing the break point by applying more number of load)

Capacity testing:-

- Testing as requested the application is accepting the load.

Reliability testing;-

- How quickly the application will get recover from the defect
- Reliability of an application due to heavy load.

Scalability testing:

- Able to deal with condition when the load increases automatically
- The system can handle the increase in user traffic, data volume….